Extreme Events and Stock Market Behavior: Why Crashes Differ, 2000–2026

DATA 400 · Jules Dao Vu Anh · October 26, 2026

Abstract

This study asks whether the cause of a stock market crash helps explain how deep it goes, how long the market takes to recover and which assets protect investors. Using daily data from 2000 to 2026, it identifies extreme days in the S&P 500 with volatility-adjusted rules, groups them into 17 crises and classifies each crisis by the economic channel through which it reached prices: expected cash flows, the risk-free interest rate, the risk premium, or the unwinding of an overvaluation.

Recovery times differed sharply by type. Fear-driven sudden events regained their pre-crisis level in a median of 10 trading days, growth and credit crises in 77 and rate and inflation shocks in 115; the two valuation unwinds took 320 and 1,472 days. Crisis type remained associated with recovery time after controlling for the depth of the fall, although with 17 crises this result is exploratory. Treasury bonds rose in every growth and credit crisis but in only one of the three rate and inflation shocks, and in 2022 they fell together with stocks. In a 60/40 portfolio, most of the protection came from holding fewer stocks rather than from bond gains. Scenarios built from past crises of the same type did not predict portfolio losses more accurately than a generic crash scenario. The cause of a crash is therefore useful for interpreting it and for designing stress tests, but not for forecasting its size.

1. Introduction and research question

The 2008 financial crisis and the 2020 COVID crash both cut the S&P 500 by more than 30%. After the 2009 trough the index needed almost four years to regain its pre-crisis level; after the 2020 trough it needed about four and a half months. In 2008 Treasury bonds cushioned a balanced portfolio; in 2022 they added to its losses. The size of a fall alone does not tell an investor what to expect.

> **How do different types of extreme events move the U.S. stock market, why do some crashes recover quickly while others take years, and what does this mean for protecting a portfolio?**

1.1 Why stock prices fall: three channels

A simple valuation formula organises the analysis. The value of the market equals expected future cash flows discounted at a rate made up of the risk-free interest rate and a risk premium, where g is the expected growth of cash flows:

$$P = \frac{E[\text{cash flows}]}{\text{discount rate} - g}, \qquad \text{discount rate} = r_f + \text{risk premium}$$

A large fall must come through at least one of three channels, and each implies different behavior in other markets:

Channel	What changes	Typical trigger	What else should happen	Expected recovery
Cash flows	Expected profits fall	Recession, credit crisis, pandemic	Yields fall as investors seek safety and expect rate cuts	Slow, unless policy replaces lost income

Campbell and Vuolteenaho (2004) split stock returns into news about cash flows and news about discount rates, and show that cash-flow news has a permanent effect on prices, while discount-rate news is partly reversed by higher returns later. If the same holds in crises, the channel through which a crisis works should shape how deep it goes, how quickly prices recover and which assets protect investors.

A fourth pattern, the end of an overvaluation, fits the discount-rate channel in reverse. Cochrane (2011) shows that most of the variation in market valuation ratios reflects changing expected returns rather than changing expected cash flows, so an unusually high price signals an unusually low risk premium. When such a market falls, the risk premium is returning to normal rather than spiking, and the pre-crash price is not a level the market should be expected to regain quickly. Greenwood, Shleifer and You (2019) show that sharp price run-ups in an industry substantially raise the probability of a later crash, which makes these episodes a recognisable category. This study treats them as a separate family, called valuation unwinds.

1.2 Hypotheses

1. **H1:** Crises that damage expected cash flows, raise interest rates or unwind an overvaluation fall deeper and take longer to recover than fear-driven sudden events, even after allowing for the depth of the fall.
2. **H2:** Treasury bonds hedge stocks in growth and credit crises but not in rate and inflation shocks.
3. **H3:** Defensive sectors fall less than cyclical sectors, and the worst-hit sector reflects the channel of the crisis.
4. **H4:** Recovery is faster when monetary policy is able to ease.
5. **H5:** Stress scenarios built from past crises of the same type predict portfolio losses better than a generic crash scenario.

2. Disciplinary motivation

The question sits where financial economics, econometrics and risk management meet.

Financial economics explains why prices fall, and a body of work shows that crises of different kinds behave differently. Muir (2017), studying financial crises, wars and recessions in 14 countries over more than 140 years, finds that asset prices fall in all three, but that expected returns rise sharply only in financial crises. Reinhart and Rogoff (2009) document that after severe banking crises equity prices fall by about 55% over roughly three and a half years. Reviewing fifteen U.S. stock market crashes, Mishkin and White (2002) conclude that financial instability, rather than the crash itself, is the key problem for monetary policymakers. Bloom (2009) identifies sixteen major volatility shocks in U.S. stock market history, many of them political or military events such as 9/11 and the two Gulf wars. Together these studies suggest that classifying crises by their cause is economically meaningful.

Econometrics supplies the tools to define “extreme” when volatility itself changes over time (J.P. Morgan/Reuters, 1996; Lee and Mykland, 2008; Longin, 1996). Risk management asks the practical question: since correlations between markets rise in falling markets (Longin and Solnik, 2001), which protections still work when they are needed?

The question is also current. In 2022 stocks and bonds fell together for most of the year, an outcome the previous two decades had made seem unlikely. Campbell, Sunderam and Viceira (2017) show that the co-movement of Treasury bonds with stocks has changed over time, from unusually high in the early 1980s

to negative in the 2000s. Understanding why 2022 happened, and whether it could have been anticipated, motivates this study.

3. Data

Daily data cover 6,737 trading days from January 3, 2000 to October 15, 2026. An extra year of data from 1999 is used only to start the volatility estimate in Section 4.1.

Data	Source	Used for
S&P 500 price index	Yahoo Finance	Finding crises; measuring depth and recovery
S&P 500 index fund (VFINX), total return	Yahoo Finance	Portfolio equity holding
Intermediate and long Treasury funds (VFITX, VUSTX)	Yahoo Finance	Bond hedges
Gold and crude oil futures, U.S. dollar index	Yahoo Finance	Safe havens and commodity shocks
VIX, 10-year and 3-month Treasury yields	Yahoo Finance	Fear, interest rates, policy stance
Nine Select Sector SPDR funds	Yahoo Finance	Sector behavior
Baa credit spread, CPI, official Treasury yields	FRED	Credit stress, inflation, checking the yield data

Two choices need explaining. Crisis depth and recovery are measured on the S&P 500 price index, the standard way to date market declines, while portfolios use total-return funds, because an investor's experience includes dividends. Total-return figures are therefore slightly better than price-index figures for the same crisis. The yields from Yahoo Finance were checked against the official FRED series, with a median daily difference of 0.01 percentage points. Oil's negative settlement price on April 20, 2020 was removed; oil is used only as a market indicator, never as a portfolio holding.

The raw data are not redistributed, because Yahoo Finance's terms do not permit it. The accompanying notebook downloads the same fixed date range, so every result in this report can be reproduced.

4. Method

4.1 Identifying extreme days

A 2.5% daily fall is routine in late 2008 and rare in 2017, so a fixed threshold alone finds too many crisis days in volatile years and too few in calm ones. Each day's return is therefore also compared with the volatility the market expected that day, estimated only from earlier returns with an exponentially weighted moving average:

$$\sigma_t^2 = \lambda\,\sigma_{t-1}^2 + (1-\lambda)\,r_{t-1}^2, \qquad z_t = \frac{r_t}{\sigma_t}, \qquad \lambda = 0.94$$

Here r is the daily log return, σ is the volatility expected for that day and z is the day's move measured in units of that volatility. The weight $\lambda = 0.94$ is the RiskMetrics standard for daily data and gives recent days more influence than distant ones (J.P. Morgan/Reuters, 1996). A day was flagged as extreme if it met any of three rules:

- **Rule A, surprise:** $|z| > 3$, a move more than three times the expected volatility. Scaling each return by an estimate of current volatility follows Lee and Mykland (2008).

- **Rule B, size:** a daily move larger than 2.5% in either direction, the threshold Baker, Bloom, Davis and Sammon (2021) use to define stock market jumps.
- **Rule C, rarity:** the day is among the largest 1% of absolute moves in the sample (4.26% or more), in the spirit of extreme-value analysis of return tails (Longin, 1996). Every such day also meets Rule B, so Rule C adds no extra days; its role is to identify the most severe episodes (Section 4.2). Because its cut-off uses the whole sample, it is retrospective; a version using only past data is tested in Section 4.6.

As an independent check, monthly volatility was compared with its long-run trend following Bloom (2009): a month is a volatility shock if it is more than 1.65 standard deviations above a Hodrick–Prescott trend. All twelve such months fall inside crises on the final list.

4.2 From extreme days to 17 crises

Crises produce clusters of extreme days, so flagged days fewer than 20 trading days (about one calendar month) apart were merged into one episode. This produced 73 episodes, including rallies and short-lived selloffs. Each episode was labelled from contemporary news coverage with what happened and the channel through which it reached prices. This follows Baker et al. (2021), who classify large daily moves by the cause reported in next-day newspapers, and the earlier work of Cutler, Poterba and Summers (1989), who examine the news on the days of the largest market moves: the data choose the dates, and news reports explain them.

Episodes were turned into crises in three steps:

1. **Selection rule.** An episode had to be included if at least one of its days was flagged by all three rules (12 such “core” episodes), or if the index fell 10% or more from its 60-day high, the conventional threshold for a market correction. The notebook confirms that no episode meeting either condition is left out.
2. **Grouping.** Episodes belonging to the same crisis were combined; the 2007–09 financial crisis, for example, produced seven separate episodes. This gives 13 crises selected by rule.
3. **Contrast cases.** Four smaller, clearly news-driven events were added by judgement: the Shanghai crash spillover (2007), the Brexit vote (2016), the yen carry-trade unwind (2024) and the start of the Iran war (2026). Without them the sudden-event family would contain only three crises. Because all four are small sudden events, this choice could flatter that family’s fast recoveries, so Section 5.1 tests whether the results survive without them.

For three crises the market slid before any single day crossed a threshold, so the event date is the news date instead of the first flagged day: the yuan devaluation (August 11, 2015), the Federal Reserve minutes signalling faster tightening (January 5, 2022) and the first trading day after the U.S. and Israeli strikes on Iran (March 2, 2026). The appendix lists every crisis and the reason it is included.

4.3 Classifying crises by channel

Each crisis was assigned to a family before its asset behavior was examined, using one decision rule: identify the cause reported in the news, then ask which part of the valuation formula it affected first.

Family	Decision rule	Crises	Supporting literature
Growth & credit	News points to weaker earnings, recession risk, or stress in banks or sovereign debt (cash-flow channel)	2008 financial crisis; Euro crisis 2010; U.S. downgrade and euro crisis 2011; China 2015–16; COVID 2020	Financial crises bring large, persistent price falls (Reinhart and Rogoff, 2009; Muir, 2017)

Family	Decision rule	Crises	Supporting literature
Rates & inflation	News points to rising interest rates, inflation or monetary tightening (risk-free rate)	Volmageddon 2018; Q4 2018 Fed tightening; 2022 inflation bear market	The co-movement of bonds and stocks depends on the inflation environment (Campbell, Sunderam and Viceira, 2017)
Valuation unwind	A market or sector that had risen far above its fundamentals deflates, without a single outside trigger	Dot-com bust 2000–01; accounting-scandal bear market 2002	Sharp run-ups raise crash risk (Greenwood, Shleifer and You, 2019); high valuations signal low expected returns (Cochrane, 2011)
Sudden event	An outside shock (war, terror, a political vote, a foreign market crash or forced selling) raises fear while expected profits are largely unchanged (risk premium)	9/11; Iraq war 2003; Shanghai 2007; Brexit 2016; yen carry unwind 2024; tariff shock 2025; Iran war 2026	Political and military events account for many volatility shocks (Bloom, 2009) and many large daily moves (Baker et al., 2021)

Four assignments are debatable, and the reasoning is stated so that the reader can disagree:

- **Volmageddon (February 2018)** began after a strong wage report raised inflation and interest-rate fears and was then amplified by the collapse of funds that bet on low volatility. It is classed by its trigger, rising yields, as a rate shock.
- **China 2015–16** had a foreign trigger, the yuan devaluation, but markets treated it as a sign of slowing global growth, and oil and energy stocks collapsed. It is classed as a growth shock.
- **The 2025 tariff shock** raised uncertainty and was largely reversed after the tariffs were paused. It is classed as a sudden event, although tariffs also threaten inflation and profits.
- **The Iran war (2026)** is classed by its trigger as a sudden event, but it reached markets mainly through oil prices (Section 5.2).

Section 4.6 tests whether the conclusions survive reassigning these crises.

4.4 Measuring each crisis

For each crisis, the *event leg* measures the damage of the event itself:

- **Base:** the last close before the event date.
- **Trough:** the lowest close from the event date until 20 trading days after the crisis's last flagged day.
- **Depth:** the fall from the base to the trough.
- **Recovery:** the number of trading days from the trough until the index first closes at or above the base.

Recovery therefore measures an investor's waiting time to break even, not how long the original shock's effect lasted. In long episodes such as 2000–02 and 2007–09, later shocks occur inside the recovery period, so recovery times describe what followed each type of crisis, not what the original shock alone caused.

To study other markets, each crisis is split into a *fall*, from the base to the S&P 500 trough, and a *rebound*, the 60 trading days after the trough. The returns of Treasuries, gold, oil, the dollar and nine sectors were measured over both phases.

4.5 Portfolios and stress tests

Portfolio	Holdings	What it represents
S&P 500	100% S&P 500 fund	The market
60/40	60% S&P 500, 40% intermediate Treasuries	The classic balanced portfolio of many retirement savers
Tech tilt	50% S&P 500, 50% technology sector	A growth-heavy investor
Defensive	One third each in consumer staples, health care and utilities	An investor who avoids economically sensitive sectors
Diversified	40% S&P 500, 30% long Treasuries, 15% intermediate Treasuries, 15% gold	A multi-asset portfolio that relies on several hedges

Every portfolio is bought at the base date and held without rebalancing or trading costs, so its weights drift with prices. Five tests were applied:

1. **Historical replay:** each portfolio's return over each crisis's fall, and its path for up to five years afterwards.
2. **Loss attribution:** the 60/40 portfolio's advantage over the S&P 500 is split into two parts. Holding 40% less stock reduces the loss by 0.4 times the S&P 500's fall, and the bonds add 0.4 times their own return. In 2008 the S&P 500 fund fell 52.3% and the 60/40 portfolio 23.9%: holding less stock saved $0.4 \times 52.3 = 20.9$ points, and bond gains saved the remaining 7.5.
3. **Hypothetical scenarios:** five forward-looking scenarios (Section 5.5) specify moves in stocks, the 10-year yield, oil and gold, and translate them into portfolio returns. Each sector moves according to its sensitivity to the S&P 500 and to oil, estimated only on high-fear days (when the previous day's VIX closed above 25), because co-movement rises in stress (Longin and Solnik, 2001). Treasury funds move according to their estimated sensitivity to the 10-year yield: about 4.9% per percentage point for intermediate and 12.0% for long Treasuries. Gold moves as the scenario specifies.
4. **Prediction tests (H5):** each crisis was predicted from other crises in two ways. A *family scenario* applies the median asset moves of crises in the same family; a *generic scenario* applies the median moves of all crises, whatever their type. In the walk-forward design, only crises that occurred earlier are used; in the leave-one-out design, all other crises are used.
5. **Value-at-Risk (VaR):** for each portfolio and crisis, a standard parametric 99% VaR was estimated from the portfolio's daily volatility in the year before the event and scaled to the length of the fall: $\text{VaR} = 2.33 \times \sigma \times \sqrt{h}$, where h is the number of trading days in the fall.

4.6 Robustness checks

- **Detection:** the crisis list was rebuilt from scratch under 57 alternative settings of the volatility weight, the cut-offs and the merge gap, including versions that drop rules or replace Rule C with a past-data-only cut-off. Every setting found at least 15 of the 17 crises, and the main and past-data-only settings found all 17.
- **Classification:** the main conclusions were recomputed under five alternative classifications: the Iran war or the 2025 tariffs as rate shocks, Volmageddon as a sudden event, all three changes together, and the overlapping 2000–03 crises merged into one.
- **Influence of single crises:** key statistics were recomputed after dropping each crisis in turn.
- **Selection:** results for H1 were recomputed without the four crises added by judgement (Section 5.1).

5. Findings

5.1 H1: crisis type is associated with depth and recovery

Family	Crises	Median fall	Median days to recover	Range of recovery (days)
Sudden event	7	−7.8%	10	8 to 28
Growth & credit	5	−14.6%	77	14 to 971
Rates & inflation	3	−18.4%	115	70 to 318
Valuation unwind	2	−27.8%	896	320 and 1,472

S&P 500 price index, event leg. All 17 crises had recovered by October 2026. On a total-return basis recoveries are shorter, but the ordering of families is the same.

The ordering is consistent with the channel framework. A sudden event raises the risk premium; once the uncertainty clears, expected profits are largely unchanged, and prices returned within weeks. A growth or credit crisis damages expected earnings, and recovery took months. Rate shocks recovered more slowly than growth and credit crises of similar size, consistent with a central bank that is fighting inflation and therefore cannot cut rates to support prices. Valuation unwinds took years, consistent with prices returning to fundamentals rather than to a pre-crash level that was itself inflated.

The VIX, a measure of expected stock market volatility, tells the same story. It rose a median 2.1 points for each 1% fall in growth and credit crises and 1.6 points in sudden events, but only 0.6 points in valuation unwinds: bubbles deflated slowly and without panic.

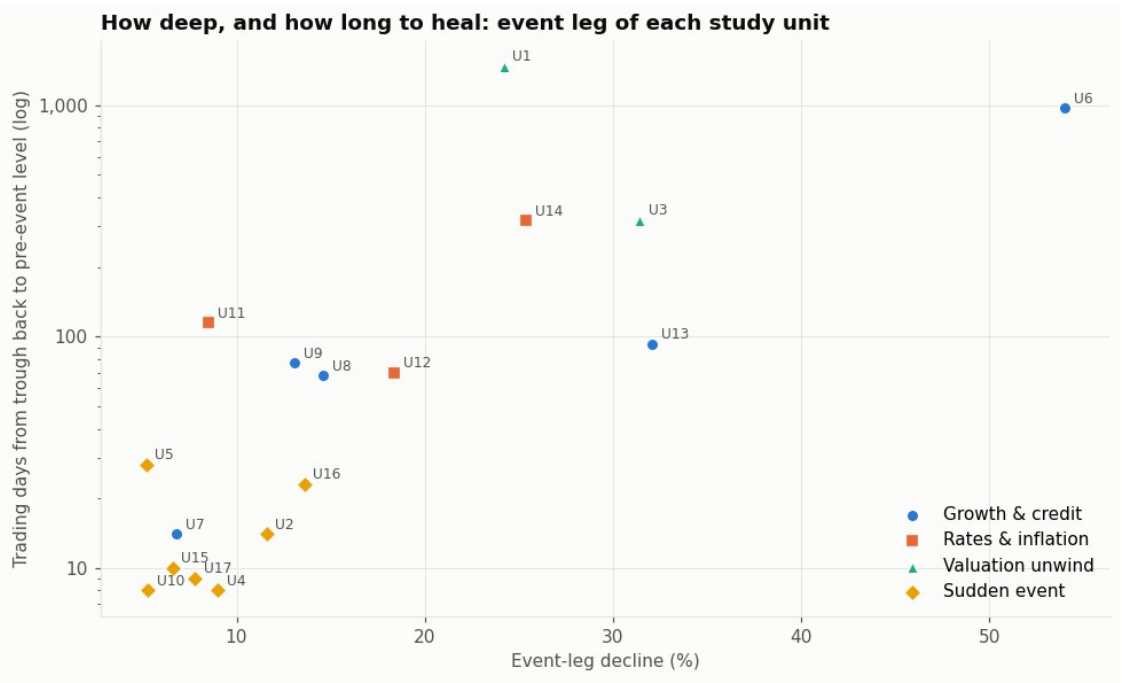


Figure 1. Event-leg decline against trading days to recover (log scale), by shock family. Point labels U1–U17 match the crisis numbers in the appendix.

In Figure 1 the families occupy distinct regions: sudden events in the lower left, valuation unwinds at the top, and growth and credit crises along a diagonal from the 2010 Euro crisis (−6.8%, 14 days) to 2008 (−54.0%, 971 days). Two crises stand out. COVID fell 32% yet recovered in 93 trading days, while

monetary and fiscal support was unprecedented. Volmageddon fell only 8.5% but needed 115 days, while the Federal Reserve was still raising rates. Both are consistent with a role for policy (Section 5.4), although two cases cannot establish it.

Is it the type of shock, or only the size of the fall? Families differ in depth as well as type, and deeper falls take longer to recover. To separate the two, the logarithm of recovery time was regressed on the logarithm of the decline and on family indicators, with sudden events as the baseline:

Model	R^2 (log recovery time)
Depth only	0.68
Family only	0.71
Depth and family	0.84

Effect in the depth-and-family model	Estimate
Depth: decline 1% larger in relative terms (e.g., 20% → 20.2%)	about 1.2% longer recovery
Growth & credit vs. sudden event, same depth	about 2.6 times as long
Rates & inflation vs. sudden event, same depth	about 4.7 times as long
Valuation unwind vs. sudden event, same depth	about 12 times as long
Family terms jointly, Freedman–Lane permutation test (20,000 draws)	p = 0.035
Same test, dropping each crisis in turn	p from 0.009 to 0.095; below 0.05 in 7 of 17

With 17 crises, conventional standard errors are unreliable, so significance was judged with the permutation test of Freedman and Lane (1983). It shuffles the residuals of the depth-only model, which keeps the relationship between depth and recovery intact, and asks how often the family terms then add as much explanatory power as they do in the real data. A simpler test that shuffles the family labels gives the same p-value.

Does the result depend on the crises added by judgement? The table below repeats the analysis under stricter selection rules. These figures are computed directly from the crisis-level values in the appendix.

Sample	Crises	Sudden event	Growth & credit	Rates & inflation	Valuation unwind	Family effect, p
All crises (main analysis)	17	10	77	115	896	0.035
Selected by rule only	13	14	77	115	896	0.053
Event-leg fall of 10% or more	10	18	85	194	896	0.23

Median trading days to recover, and the Freedman–Lane p-value for the family terms given depth.

The ordering of families is the same in every sample. Without the four contrast cases, the family effects at the same depth are, if anything, larger (3.2, 6.0 and 13 times as long as a sudden event), but the p-value rises to 0.053. With only crises that fell at least 10%, two families have only two members each, too few for a meaningful test. The evidence therefore supports H1 as a consistent pattern whose statistical significance depends on sample size. The case for H1 rests mainly on the size and consistency of the differences, with the regression as supporting evidence.

5.2 H2: Treasuries hedged growth shocks, not rate shocks

Family	Intermediate Treasuries gained	Gold gained	Stock–bond correlation in falls
Growth & credit	5 of 5	4 of 5	−0.52
Sudden event	6 of 7	5 of 7	−0.38
Valuation unwind	2 of 2	1 of 1	−0.32
Rates & inflation	1 of 3	1 of 3	+0.05

"Gained" counts crises in which the asset rose over the fall. The correlation is between daily returns of the S&P 500 fund and the intermediate Treasury fund, pooled over all days in the family's falls. Gold data begin in August 2000, so the dot-com bust has no gold observation.

The valuation formula explains the pattern. In a growth crisis, investors expect weaker earnings and lower interest rates: stocks fall through the cash-flow channel while bond prices rise. In a rate shock, rising interest rates are themselves the cause of the stock decline, and the same rising rates push bond prices down. In 2022 the S&P 500 fund fell 24.5% (a 25.4% fall in the price index), intermediate Treasuries fell 11.1% and long Treasuries 27.9%.

Only three rate shocks are available, so a formal test is weak: comparing the one rate shock in three in which Treasuries gained with the 13 of 14 other crises, a one-sided Fisher exact test gives $p = 0.063$. H2 is nevertheless the study's most consistent result, for three reasons. The same split appears in three different measures (how often bonds gained, the correlation inside the falls and the long-run correlation in Figure 2). It follows directly from the valuation formula. And it matches the findings of Campbell, Sunderam and Viceira (2017) for a much longer sample.

A natural concern is that choosing crash windows creates the negative correlation. Using a rule fixed in advance instead (days when the previous day's VIX closed above 25), the stock–bond correlation was −0.38, against −0.19 on other days. This matches the −0.38 found on crisis days, so the hedge is not an artefact of how the windows were chosen.

In this sample, oil moved with the channel. It fell sharply in the demand-driven crises (−39% in 2008 and −57% during COVID), when bonds protected, and rose in 2022 (+13%) and the Iran war (+54%), when they did not. Falling oil is consistent with weaker demand and expected rate cuts; rising oil with inflation that keeps rates high. The Iran war shows why the trigger and the channel must be kept apart. Its trigger was a sudden geopolitical event, and its quick recovery fits that family, but it reached markets through oil and inflation expectations, and Treasuries and gold both fell, as in a rate shock.

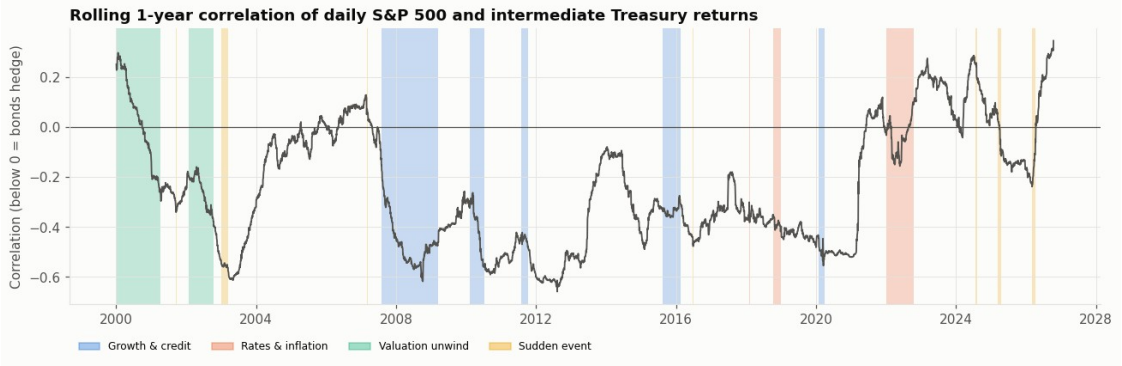


Figure 2. Rolling one-year correlation of daily S&P 500 and intermediate Treasury returns, with crises shaded by family. Below zero, bonds tend to rise when stocks fall.

Figure 2 shows that the stock–bond relationship is a regime, not a constant. The correlation was negative for most of 2001–2020, reaching about −0.6 around 2008, the period that built the reputation of the 60/40 portfolio. It turned positive in 2021 as inflation returned and has mostly stayed positive since. Both positive periods, around 2000 and after 2021, were times when inflation and interest rates were prominent concerns, which is consistent with the explanation of Campbell, Sunderam and Viceira (2017). A portfolio built on the relationships of 2002–2020 was built for a regime that ended in 2021.

5.3 H3: defensive sectors cushioned the fall, and the worst sector reflected the channel

Defensive sectors (consumer staples, health care and utilities) sell goods and services that people buy in any economy, so their profits move less in downturns. Their equal-weighted return was compared with that of the six cyclical sectors (technology, financials, energy, consumer discretionary, industrials and materials). Defensives beat cyclicals during the fall in every family, by a median 11.2 percentage points in growth and credit crises, 12.6 in valuation unwinds, 4.9 in sudden events and 4.6 in rate shocks. Cyclicals then led the rebound by 4 to 7 points over the following 60 trading days.

The worst-performing sector often pointed to the channel: financials in the 2008 banking crisis, energy in growth scares accompanied by an oil collapse, technology in the valuation unwinds, and consumer discretionary in 2022. In the dot-com bust technology fell 60% while utilities, energy, financials and staples all rose: money rotated between sectors rather than leaving the market. In 2008, by contrast, every sector fell, because a banking crisis affects the whole economy. Sector diversification therefore helped most when the shock was concentrated in one part of the economy, and least when it was system-wide.

5.4 H4: policy easing is associated with faster recovery, weakly

The 3-month Treasury bill yield closely follows the expected policy rate, so a fall in it is used as a measure of monetary easing. In the five crises where it fell by more than 0.5 percentage points within six months of the event, the median recovery took 28 trading days, against 69 in the other twelve. The mechanism is plausible: Bernanke and Kuttner (2005) find that an unexpected 25-basis-point cut in the federal funds rate target is associated with a rise of about 1% in broad stock indexes. The evidence here is weak, however, because causation runs both ways: the Federal Reserve eases most in the worst crises, and the 2008 recovery still took 971 days.

Conditions before each crisis (the VIX, the level of interest rates, the yield curve, the previous year’s return, the credit spread and inflation) showed no strong relationship with depth or recovery: no rank

correlation exceeded 0.33 in absolute value. In this sample, the state of the market before a crisis said little about how severe it would be.

5.5 Stress test: which portfolio wins depends on the shock

Median portfolio return over the S&P 500's fall, by family:

Family	S&P 500	60/40	Tech tilt	Defensive	Diversified
Growth & credit	−14.3%	−7.1%	−12.2%	−5.8%	+0.5%
Rates & inflation	−18.1%	−9.7%	−19.4%	−9.1%	−4.2%
Sudden event	−7.7%	−4.4%	−8.3%	−7.2%	−2.6%
Valuation unwind	−26.9%	−9.7%	−41.4%	−8.7%	−3.1%

Total-return funds, so the S&P 500 figures differ slightly from the price-index falls in Section 5.1.

The medians hide the most important case. The Diversified portfolio gained in the typical growth crisis yet lost 21.0% in 2022, close to the S&P 500's 24.5%, because 45% of it was in Treasuries that fell with stocks. The 60/40 portfolio cut the 2008 loss from 52% to 24%, but in 2022 it lost 19% against 24.5%.

Where the protection came from. Using the attribution in Section 4.5, most of the 60/40 portfolio's advantage came from holding fewer stocks rather than from hedging. The share of its stock loss offset by bond gains was a median 21% in growth and credit crises, 42% in valuation unwinds, 13% in sudden events and 0% in rate shocks. The more hedges a portfolio holds, the more its outcome depends on the type of shock: the Diversified portfolio's hedges more than offset its stock loss in five crises, but in 2022 they more than doubled it.

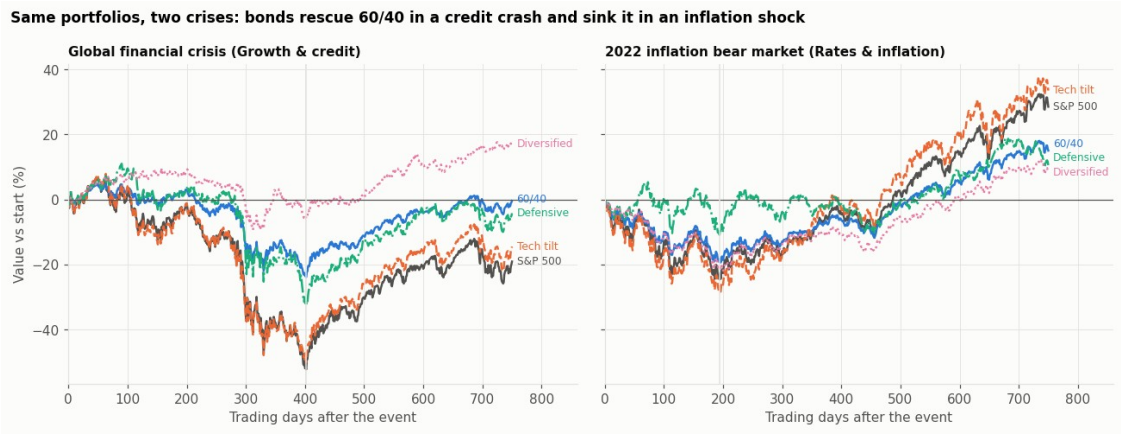


Figure 3. Value of each portfolio relative to its starting value, in trading days after the event: the 2008 financial crisis (left) and the 2022 inflation bear market (right).

Figure 3 puts the two regimes side by side. In 2008 the portfolios fan out: around the trough the S&P 500 was down about 52% and Diversified less than 10%. In 2022 they fall together, and only Defensive stays near break-even. Three years after 2022 began, the order had reversed: the S&P 500 and Tech tilt were up about 30% to 35%, while Diversified, the winner of 2008, was lowest at about +9%. The same portfolio can be the best or the worst choice depending on the channel of the shock.

Looking forward. Because the sample contains only one inflationary bear market, history cannot show the full range of risks. Five hypothetical scenarios, each anchored on a real episode but more severe, were therefore applied:

Scenario (anchor)	S&P 500	10-year yield	60/40	Tech tilt	Defensive	Diversified
Deflationary bust (2008, COVID)	−35%	−1.50 pts	−18.0%	−36.9%	−22.4%	−5.2%
Dash for cash (March 2020)	−30%	+0.25 pts	−18.5%	−31.6%	−19.1%	−14.6%
Stagflation (2022 with an oil shock)	−25%	+1.50 pts	−18.0%	−26.9%	−16.7%	−16.5%
Sell America (2025 tariffs, larger)	−20%	+0.75 pts	−13.5%	−21.2%	−12.9%	−9.0%
Oil supply shock (Iran war, larger)	−12%	+0.50 pts	−8.2%	−13.4%	−8.7%	−8.5%

Oil and gold moves are also specified in each scenario: −40% and +15% (deflationary bust), −40% and −10% (dash for cash), +40% and 0% (stagflation), −10% and +15% (Sell America), +80% and −10% (oil supply shock).

Diversified is by far the most resilient portfolio in a deflationary bust but no safer than 60/40 or Defensive in stagflation, where rising yields hit its bonds. The 60/40 portfolio's worst case is a dash for cash, in which investors sell stocks and bonds together. Defensive loses more than 60/40 in a deflationary bust, because it holds no bonds and its sectors move more closely with the market on high-fear days (a combined sensitivity of about 0.65) than in calm markets.

The scenario model was checked by feeding it the market moves that actually occurred in each crisis. With sensitivities estimated only from data before each crisis, it reproduced realised portfolio losses with an average error of 1.5 percentage points across 16 crises (1.2 points with full-sample sensitivities). This tests how well the model turns market moves into portfolio losses; it does not test whether the scenario moves themselves are realistic, which remain assumptions.

5.6 H5: knowing the crisis type did not improve prediction

Validation design	Crises scored	Mean error, family scenario	Mean error, generic scenario	Family scenario closer
Walk-forward, level	12	7.8 pts	6.5 pts	6 of 12
Walk-forward, conditional on depth	12	2.6 pts	2.6 pts	8 of 12
Leave-one-out, level	17	7.9 pts	7.1 pts	9 of 17
Leave-one-out, conditional on depth	17	3.9 pts	3.3 pts	9 of 17

Mean absolute error per crisis, in percentage points of portfolio return. The level test predicts each portfolio's return directly. The conditional test is given the crisis's actual S&P 500 fall and predicts only each portfolio's return relative to it. Walk-forward starts once three earlier crises are available, and a crisis is scored only if an earlier crisis of its family exists.

Family scenarios were closer than the generic scenario in between half and two-thirds of crises, with similar or larger average errors, so they were not reliably better and H5 is not supported. Both designs are also generous to family scenarios: they assume the crisis's family is known when it begins, using the label assigned with hindsight, and the conditional test uses the actual market fall. An investor would first have to identify the family as the crisis unfolds, which this study does not test.

The crisis-level results show why. Family scenarios worked for sudden events, which are numerous and alike, but failed when one extreme crisis dominated a family's history: the 2010 Euro crisis, predicted from the 2008 crisis alone, missed by 29.8 points against 8.6 for the generic scenario. They also failed when the regime changed, since no rate shock before 2022 involved bonds falling with stocks. Most of the error came from how deep a crisis would go, which no historical average anticipates.

5.7 A related finding: calm-market risk estimates

Separately from H5, the VaR test asks whether a standard risk model, calibrated in the year before each crisis, anticipated the loss. The ratio of the actual loss to the 99% VaR had a median of 1.27 in rate shocks and 1.20 in sudden events, meaning losses exceeded the model's "one-in-a-hundred" bound. In growth and credit crises and valuation unwinds the median ratio was 0.60: these slower falls stayed within the bound, because scaling volatility by the square root of a long fall makes the bound very wide. Since these windows were selected as crises, the results do not measure how often VaR fails in general. They show that fast shocks arriving after calm periods are the ones that calm-market risk estimates understate most.

5.8 Summary of the hypotheses

Hypothesis	Verdict	Main evidence
H1: type matters beyond depth	Supported as an exploratory association	Consistent ordering of recovery times in every sample; family effect $p = 0.035$, weaker in smaller samples
H2: Treasuries hedge growth, not rate shocks	Supported; the most consistent result	Treasuries gained in 13 of 14 non-rate crises but 1 of 3 rate shocks; correlation and regime evidence agree
H3: defensives cushion; worst sector reveals channel	First part supported; second part partly	Defensives beat cyclicals in every family; the worst sector matched the channel in several crises but not all
H4: easing speeds recovery	Weak support	28 vs. 69 days, but causation runs both ways
H5: family scenarios predict better	Not supported	Family scenarios no more accurate than a generic crash scenario

6. What the crises suggest about how markets work

Read together, the findings suggest five lessons about market behavior. They are interpretations of 17 crises rather than proven laws.

1. A crash is a repricing, and its cause helps explain what follows. Every stock price reflects expected cash flows and a discount rate. When only fear rose, it faded and prices returned quickly: 9/11 cut the index 11.6% and was recovered in 14 trading days. When expected profits fell, recovery waited for the economy: the 2011 downgrade crisis, only slightly deeper at 14.6%, took 68 days. When prices had been too high to begin with, the old level was itself the problem: after the dot-com bust it took 1,472 trading days to return. Falls of similar size can lead to very different waits.

2. A safe asset is safe only against a particular danger. Over a few weeks, most of a Treasury fund's return comes from changes in interest rates, so holding Treasuries in a crash works largely like a position on rates falling. That position paid off when the danger was weak growth, because rates fell, and lost when the danger was inflation, because rates rose. Baur and Lucey (2010) find that gold acts as a safe haven in extreme stock market conditions, but in this sample its behavior also depended on the shock: it rose 38% in 2008 but fell in 2022 and in the Iran war, when rate cuts became less likely.

3. Correlations reflect what investors fear. The stock–bond correlation was negative for two decades and turned positive when inflation returned. Campbell, Sunderam and Viceira (2017) link such shifts to

how inflation moves with the real economy. Portfolio models that estimate correlations from recent history implicitly assume that the market will fear the same things tomorrow as yesterday, and 2022 shows how costly that assumption can be.

4. Policy is part of the price, but not a guarantee. Recoveries were faster on average when the central bank could ease, and an unexpected cut tends to lift stock prices (Bernanke and Kuttner, 2005). The evidence for this in the sample is weak, however, and the fastest recoveries of all, after sudden events, mostly needed no easing. What the data do suggest is that inflation shocks are doubly damaging: they raise discount rates directly and limit the policy support that often helps end a crisis.

5. Explaining is easier than forecasting. The framework that explains each crisis after the fact did not forecast portfolio losses better than a generic crash scenario, and conditions before each crisis said little about its severity. The most promising practical use is diagnostic: early signals such as the direction of oil prices, the move in Treasury yields and which sectors lead the decline may indicate a crisis's channel, and therefore which protections are likely to work. This study shows that these signals lined up with the channels after the fact; whether they identify the channel reliably in the first days of a crisis remains a hypothesis for future testing.

7. Stakeholders

Stakeholder	What the study shows	Implication
Individual and retirement investors	A 60/40 portfolio protects mainly by holding fewer stocks; bonds did not protect in rate and inflation shocks	Choose equity exposure for the loss you can bear, rather than relying on bonds to offset it
Asset managers and financial advisers	The value of hedges depends on the channel of the shock	Report portfolio risk by scenario type, not as a single number
Risk managers	History-based scenarios miss new regimes, and calm-period VaR understates fast shocks	Combine historical replay with hypothetical scenarios such as stagflation and a dash for cash
Pension funds and endowments	Recoveries ranged from two weeks to almost six years	Plan liquidity for long drawdowns driven by valuation or rates
Regulators and central banks	Recovery is associated with easing, and inflation limits that option	Include scenarios in which stocks and bonds fall together

8. Limitations

- **Small sample.** Seventeen crises in four families, with two to seven per family, can describe patterns but not establish them. The p-values near 0.05 move when single crises are dropped.
- **Selection and judgement.** Thirteen crises were selected by rule and four were added by judgement, and grouping episodes into crises and assigning families involved judgement. Some crises mix channels; the 2025 tariffs and the Iran war both had inflationary elements. Section 5.1 and the reclassification tests show that the main conclusions survive these choices, but their exact size depends on them.
- **Hindsight.** Crisis windows, troughs and family labels are known only afterwards. The results describe what happened; they are not a real-time method.
- **Recovery measure.** Recovery is the time to regain the pre-event price level. In long episodes it includes later, unrelated shocks, so it describes what followed each type of crisis, not the effect of the original shock alone.

- **Many comparisons.** Five hypotheses and many family comparisons were examined on the same 17 crises, so some differences may look notable by chance. Results that agree across several measures, such as H2, deserve more weight than single comparisons.
- **Prices, not fundamentals.** The channels are inferred from how markets moved and what the news reported, not measured from earnings forecasts or survey data.
- **Simplified portfolios and scenarios.** Portfolios are buy-and-hold without rebalancing or trading costs, the scenario model scales daily sensitivities to large shocks, and scenario sizes are assumptions.
- **Scope.** U.S. markets since 2000 only. The sample contains a single inflationary bear market and none of the 1970s experience.

9. Conclusion

Extreme market events since 2000 differed according to the channel through which they reached prices. Fear-driven sudden events fell least and recovered in about two weeks; growth and credit crises took months; rate and inflation shocks recovered more slowly still; and the two valuation unwinds took years. These differences remained after allowing for the depth of the fall, although with 17 crises they are best read as consistent associations rather than established effects.

The channel was also closely associated with which assets protected an investor. Treasuries hedged growth and credit crises but fell with stocks when rising rates and inflation drove the decline, as in 2022 and, through oil, in the 2026 Iran war. Most of a balanced portfolio's protection came simply from holding fewer stocks.

Knowing the type of a crisis did not predict portfolio losses better than a generic crash scenario, so the framework is a tool for interpretation and stress-test design rather than forecasting. Three practical lessons follow:

1. Ask why the market is falling before deciding what will protect a portfolio.
2. Do not assume that bonds will offset stock losses; in inflation shocks they did not.
3. Test portfolios against scenarios that history has not yet shown, because the next crisis may belong to a new regime.

A longer sample that includes the inflationary 1970s, other national markets and direct measures of earnings expectations would test these conclusions further, as would a real-time test of whether a crisis's channel can be identified as it begins.

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Data and event sources

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Appendix. The 17 crises

#	Crisis	Family	Event date	Included by	Fall	Days to recover
U1	Dot-com bust	Valuation unwind	2000-01-04	Rule: core	−24.2%	1,472
U2	September 11 attacks	Sudden event	2001-09-17	Rule: core	−11.6%	14
U3	Accounting-scandal bear market	Valuation unwind	2002-01-29	Rule: core	−31.4%	320
U4	Iraq war	Sudden event	2003-01-02	Rule: 10% fall	−9.0%	8
U5	Shanghai crash spillover	Sudden event	2007-02-27	Judgement	−5.2%	28

#	Crisis	Family	Event date	Included by	Fall	Days to recover
U6	Global financial crisis	Growth & credit	2007-08-03	Rule: core	−54.0%	971
U7	Euro crisis 2010	Growth & credit	2010-02-04	Rule: core	−6.8%	14
U8	U.S. downgrade and euro crisis 2011	Growth & credit	2011-08-02	Rule: core	−14.6%	68
U9	China 2015–16	Growth & credit	2015-08-11	Rule: 10% fall	−13.1%	77
U10	Brexit vote	Sudden event	2016-06-24	Judgement	−5.3%	8
U11	Volmageddon	Rates & inflation	2018-02-02	Rule: 10% fall	−8.5%	115
U12	Q4 2018 Fed tightening and trade war	Rates & inflation	2018-10-10	Rule: core	−18.4%	70
U13	COVID-19	Growth & credit	2020-01-27	Rule: core	−32.1%	93
U14	2022 inflation bear market	Rates & inflation	2022-01-05	Rule: core	−25.4%	318
U15	Yen carry-trade unwind	Sudden event	2024-07-24	Judgement	−6.6%	10
U16	2025 tariff shock	Sudden event	2025-03-10	Rule: core	−13.6%	23
U17	Iran war	Sudden event	2026-03-02	Judgement	−7.8%	9

“Rule: core” = at least one day flagged by all three detection rules; “Rule: 10% fall” = the index fell at least 10% from its 60-day high during the episode, a screening measure that can exceed the event-leg fall shown here (for example, the Iraq war episode includes the decline before the invasion); “Judgement” = added as a contrast case (Section 4.2). Fall = event-leg fall on the S&P 500 price index; days to recover = trading days from the trough back to the pre-event level. The accompanying notebook, extreme_events_full.ipynb, reproduces every result with Restart and run all.